

Session 12 Quiz - Advanced Prompting Techniques

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:57293]

When you ask an AI a question without providing any examples or special setup, you are using the _____ technique.

- A. Chain-of-Thought
- B. Zero-Shot
- C. Few-Shot
- D. Role-Play

Correct Answer: B. Zero-Shot

Explanation: Zero-shot prompting involves asking the AI directly for a task it already knows how to do without extra guidance or examples. It is the default way of asking where no previous training or examples are given in the prompt. Chain-of-Thought, Few-Shot, and Role-Play all add additional structure.

Q2 [Type: MCQ | Difficulty: Easy | QID:57295]

Providing a pattern for the AI to follow, like showing it how to name two pets before asking for a third, is called _____ prompting.

- A. Few-Shot
- B. Creative
- C. Iterative
- D. Zero-Shot

Correct Answer: A. Few-Shot

Explanation: Few-shot prompting involves providing a small number of examples (the "shots") to help the AI understand the specific style, format, or pattern you want it to follow. Iterative refers to refining prompts over time, not providing examples.

Q3 [Type: MCQ | Difficulty: Easy | QID:57296]

To use the Role-Play technique, you should start your prompt with a phrase like '_____.'

- A. List the...
- B. Search for...
- C. Define the...
- D. You are a...

Correct Answer: D. You are a...

Explanation: Telling the AI who it should be or how it should act is the core starting point for Role-Play prompting. Phrases like "You are a pirate" or "You are a teacher" assign an identity or job to the AI, shaping its tone and perspective.

Q4 [Type: MCQ | Difficulty: Easy-Medium | QID:57297]

When dealing with a tricky math problem or logic puzzle, adding the phrase '_____' helps the AI stay accurate.

- A. Give me the answer
- B. Make it short
- C. Think step by step
- D. Use big words

Correct Answer: C. Think step by step

Explanation: Asking the AI to "think step by step" triggers Chain-of-Thought reasoning, which slows the AI down so it builds its answer like a ladder. Each step helps the AI predict the next part more accurately, reducing errors in complex problems.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:57299]

Advanced techniques like Role-Play and Chain-of-Thought are meant to _____ the 5-part framework for prompts.

- A. ignore
- B. contradict
- C. build on
- D. replace

Correct Answer: C. build on

Explanation: The source material notes that these advanced tricks enhance the basic 5-part framework (Topic, Action, Context, Format, Constraints) rather than replacing or ignoring it. They are additions that make prompts more powerful.

Q6 [Type: True/False | Difficulty: Easy-Medium | QID:57300]

True or False: Few-Shot prompting is the best way to make the AI smarter and increase its overall knowledge.

- A. False
- B. True

Correct Answer: A. False

Explanation: Few-shot prompting is designed to help the AI match a specific style or pattern for a given task, not to increase its actual intelligence or knowledge. It does not teach the AI new facts; it only shows the desired format.

Q7 [Type: True/False | Difficulty: Easy | QID:57301]

True or False: You can combine multiple techniques, such as Role-Play and Chain-of-Thought, in a single prompt.

A. False

B. True

Correct Answer: B. True

Explanation: Mixing techniques is allowed and often powerful. For example, you can ask the AI to "act like a detective" (Role-Play) and also "think step by step" (Chain-of-Thought) to get a creative voice with careful logic.

Q8 [Type: True/False | Difficulty: Easy | QID:57302]

True or False: Chain-of-Thought prompting is the best choice for simple, factual questions like 'What is the capital of France?'

A. False

B. True

Correct Answer: A. False

Explanation: Simple facts are best handled with Zero-Shot prompting. Chain-of-Thought is reserved for complex logic, math, or multi-step reasoning. Adding "think step by step" to a one-word answer is unnecessary and wasteful.

Q9 [Type: True/False | Difficulty: Easy | QID:57303]

True or False: Zero-Shot prompting is the default method most people use when they first start talking to an AI.

- A. True
- B. False

Correct Answer: A. True

Explanation: Most users simply ask a question without providing examples or special setup, which is exactly Zero-Shot prompting. It is the natural default method.

Q10 [Type: True/False | Difficulty: Easy | QID:57304]

True or False: Role-Play prompting is used when you want the AI to provide information in a very specific tone or personality.

- A. True
- B. False

Correct Answer: A. True

Explanation: Role-Play allows you to inject humor, authority, or specific character voices into the AI's replies. Examples include a pirate, a wizard, a teacher, or a grandma, each giving a distinct tone and perspective.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:57305]

Which prompting technique is most like teaching someone a new card game by playing a few practice rounds first?

- A. Zero-Shot Prompting
- B. Few-Shot Prompting
- C. Role-Play Prompting
- D. Chain-of-Thought Prompting

Correct Answer: B. Few-Shot Prompting

Explanation: Practice rounds serve as the examples that set the pattern for how the game (or the prompt) should proceed. Few-shot prompting provides a small number of examples to show the AI what you want, just like demonstrating a card game before playing for real.

Q12 [Type: MCQ | Difficulty: Medium | QID:57306]

An AI is asked to calculate the end time of a movie with an intermission. Why is Chain-of-Thought the best technique for this?

- A. It makes the AI's response sound more like a movie director.
- B. It makes the answer shorter and easier to read.
- C. It allows the AI to provide examples of other movies.
- D. It prevents the AI from skipping the intermission or making calculation errors.

Correct Answer: D. It prevents the AI from skipping the intermission or making calculation errors.

Explanation: By breaking the problem into steps (movie duration, intermission length, addition, final calculation), the AI is less likely to overlook details or make arithmetic mistakes. Chain-of-Thought reduces errors in multi-step reasoning.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:57307]

You want the AI to write a review of human food from the perspective of a sassy cat. Which technique is most important here?

- A. Role-Play
- B. Chain-of-Thought
- C. Zero-Shot
- D. Few-Shot

Correct Answer: A. Role-Play

Explanation: Assigning a specific persona (a sassy cat) is necessary to get the unique voice and perspective required. Role-Play gives the AI a character to embody, which shapes the tone, vocabulary, and attitude of the response.

Q14 [Type: MCQ | Difficulty: Easy | QID:57308]

What is a 'Prompt Library' used for in the context of these lessons?

- A. To chat with other students about their homework.
- B. To download new versions of AI software.
- C. To save and organize effective prompts for future use.
- D. To find books about how computers were invented.

Correct Answer: C. To save and organize effective prompts for future use.

Explanation: The objectives include adding "awesome new prompts" to a library to build a collection of successful tools. A Prompt Library is a personal toolkit where you store your best prompts so you can reuse them later.

Q15 [Type: MCQ | Difficulty: Easy-Medium | QID:58881]

If you use the prompt 'You are a pirate. Explain how to tie a knot step by step,' which techniques are you combining?

- A. Role-Play and Chain-of-Thought
- B. Zero-Shot and Chain-of-Thought
- C. Few-Shot and Role-Play
- D. Zero-Shot and Few-Shot

Correct Answer: A. Role-Play and Chain-of-Thought

Explanation: The prompt assigns a persona ("You are a pirate") which is Role-Play, and it asks for a logical sequence of actions ("step by step") which is Chain-of-Thought. The two techniques are combined in a single prompt.

Q16 [Type: MCQ | Difficulty: Medium | QID:57309]

Why does Chain-of-Thought prompting work so well for AI?

- A. It helps the AI remember your name for future sessions.
- B. The AI predicts the next step from the previous one, leading to higher accuracy.
- C. It allows the AI to search the internet faster.
- D. It makes the AI use more emojis and fun colors.

Correct Answer: B. The AI predicts the next step from the previous one, leading to higher accuracy.

Explanation: Building an answer line by line (step by step) prevents the AI from jumping to a wrong conclusion based on incomplete logic. Each step provides context for the next, improving overall accuracy in multi-step tasks.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:57310]

Which of these is a valid example of a Few-Shot prompt for generating band names?

- A. Make a list of 5 funny band names for small pets.
- B. Here are 2 examples: My cat → 'The Meow Mixers'; My dog → 'The Barking Beats'. Now make one for my hamster.
- C. Think step by step to come up with a band name for a hamster.
- D. You are a famous rock star. Invent a band name for a hamster.

Correct Answer: B. Here are 2 examples: My cat → 'The Meow Mixers'; My dog → 'The Barking Beats'. Now make one for my hamster.

Explanation: This prompt provides specific examples (the "shots") before asking the AI to complete a new similar task. That is the defining characteristic of Few-Shot prompting. The other options use Zero-Shot, Chain-of-Thought, or Role-Play.

Q18 [Type: MCQ | Difficulty: Easy | QID:57311]

Which technique is best for 'Simple, normal tasks AI already knows how to do'?

- A. Role-Play
- B. Chain-of-Thought
- C. Few-Shot
- D. Zero-Shot

Correct Answer: D. Zero-Shot

Explanation: For tasks like translations, basic facts, or simple questions, extra examples or personas are often unnecessary. Zero-Shot prompting (just asking the question directly) is the most efficient and effective method.

Q19 [Type: MCQ | Difficulty: Medium | QID:57312]

In the 'movie duration' example for Chain-of-Thought, what was the second step the AI followed?

- A. Find out the name of the movie.
- B. Explain what an intermission is.
- C. Add the intermission time to the base length.
- D. Check what time the movie starts.

Correct Answer: C. Add the intermission time to the base length.

Explanation: After determining the movie's duration, the next logical step is to include the intermission (break) time in the total runtime. Then the AI would add that total to the start time to find the end time.

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:57314]

What is the primary reason for 'iterating' on a prompt?

- A. To refine the prompt until the AI gives the best possible result.
- B. To delete all your previous prompts.
- C. To make the AI work faster.
- D. To change the language the AI is speaking.

Correct Answer: A. To refine the prompt until the AI gives the best possible result.

Explanation: Iteration is the process of trying, failing, and adjusting your words to get better performance from the AI. Like learning to ride a bike, you try, fall, adjust, and try again until you get it right. The goal is the best possible result, not speed or language changes.